

Chapter 4: Effect Modification

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So far we have focused on the average causal effect in an entire population. Many causal questions, however, are about **subsets** of the population: the effect of your looking up in city dwellers versus visitors, say.

Whether to target the whole population or a subset depends on the goal:

- a nationwide water fluoridation program reaches every household, so the policy maker cares about the population-average effect;
- variation across subsets matters when an intervention can be targeted, or when findings must be applied to other populations.

The theme of this chapter: there is no such thing as *the* causal effect of a treatment. The effect depends on the characteristics of the population under study.

This chapter is based on Hernán and Robins (2020, chap. 4, pp. 47-60).

1 4.1 Heterogeneity of Treatment Effects (pp. 47-48)

Table 4.1 is Table 1.1 (the counterfactual outcomes in Zeus’s family) plus an indicator V for sex: $V = 1$ for women, $V = 0$ for men.

Name	V	$Y^{a=0}$	$Y^{a=1}$	Name	V	$Y^{a=0}$	$Y^{a=1}$
Rheia	1	0	1	Kronos	0	1	0
Deme- ter	1	0	0	Hades	0	0	0
Hestia	1	0	0	Posei- don	0	1	0
Hera	1	0	0	Zeus	0	0	1
Artemis	1	1	1	Apollo	0	1	0
Leto	1	0	1	Ares	0	1	1
Athena	1	1	1	Hep- haes- tus	0	0	1
Aphrodite	1	0	1	Polyphe- mus	0	0	1
Perse- phone	1	1	1	Her- mes	0	1	0
Hebe	1	1	0	Diony- sus	0	1	0

In the whole population, $\Pr[Y^{a=1} = 1] = 10/20 = 0.5$ and $\Pr[Y^{a=0} = 1] = 10/20 = 0.5$, so the causal risk ratio is 1 and the causal risk difference is 0 (Chapter 1). The book notes that calling these deities “women” and “men” is a slight abuse of terms.

Example 1.1 (Sex-specific effects in Table 4.1). **Women** ($V = 1$, first 10 rows):

- $\Pr[Y^{a=1} = 1 \mid V = 1] = 6/10 = 0.6$ and $\Pr[Y^{a=0} = 1 \mid V = 1] = 4/10 = 0.4$
- causal risk ratio $0.6/0.4 = 1.5$; causal risk difference $0.6 - 0.4 = 0.2$: transplant **increases** risk.

Men ($V = 0$, last 10 rows):

- $\Pr[Y^{a=1} = 1 \mid V = 0] = 4/10 = 0.4$ and $\Pr[Y^{a=0} = 1 \mid V = 0] = 6/10 = 0.6$
- causal risk ratio $0.4/0.6 = 2/3$; causal risk difference $0.4 - 0.6 = -0.2$: transplant **decreases** risk.

Remark 1.1 (A Null Average Can Hide Subset Effects). A population average causal effect of zero can coexist with nonzero average causal effects within some of its subsets. In Example 1.1 the two sex-specific risk differences are equal in magnitude and opposite in sign, and each sex is 50% of the population, so they cancel exactly in the population risk difference of 0.

Exact cancellation is probably rare, but heterogeneity of individual effects is often expected because individuals differ in their susceptibility to treatment. The exception is the **sharp causal null hypothesis**: if the effect is null for every individual, the average effect in every subset is null too, and there is no heterogeneity.

1.1 Effect modification

Definition 1.1 (Effect modifier). V is a **modifier** of the effect of A on Y when the average causal effect of A on Y varies across levels of V . For binary V :

- **additive** effect modification: $E[Y^{a=1} - Y^{a=0} | V = 1] \neq E[Y^{a=1} - Y^{a=0} | V = 0]$
- **multiplicative** effect modification: $\frac{E[Y^{a=1} | V = 1]}{E[Y^{a=0} | V = 1]} \neq \frac{E[Y^{a=1} | V = 0]}{E[Y^{a=0} | V = 0]}$

Only variables V not affected by treatment A are considered as effect modifiers.

Definition 1.2 (Qualitative Effect Modification). Let V be binary with $\Pr[V = v] > 0$ for $v = 0, 1$, and let $RD_v \stackrel{\text{def}}{=} E[Y^{a=1} - Y^{a=0} | V = v]$ be the average causal effect on the additive scale in stratum $V = v$. We call V a **qualitative** effect modifier if the two stratum-specific effects point in **opposite directions**: either $RD_1 > 0 > RD_0$ or $RD_1 < 0 < RD_0$.

Example 1.2 (Sex as a Qualitative Modifier in Table 4.1). In Example 1.1, $RD_1 = 0.2 > 0$ in women and $RD_0 = -0.2 < 0$ in men, so sex is a qualitative modifier of the effect of transplant on death. It modifies the effect on both scales: the risk differences are 0.2 and -0.2 , and the risk ratios are 1.5 and $2/3$.

The book does not consider effect modification on the odds ratio scale, because the odds ratio is rarely, if ever, the parameter of interest for causal inference (see also Fine Point 4.3). Section 6.6 gives a structural classification of effect modifiers.

1.2 Effect modification depends on the scale

Proposition 1.1 (Opposite Directions Show Up on Both Scales). *Let V be binary with $\Pr[V = v] > 0$ for $v = 0, 1$, let $RD_v \stackrel{\text{def}}{=} E[Y^{a=1} - Y^{a=0} | V = v]$ be the average causal effect on the additive scale in stratum $V = v$, let $p_v \stackrel{\text{def}}{=} E[Y^{a=0} | V = v]$ be the counterfactual mean under no treatment in stratum $V = v$, with $p_v > 0$ for $v = 0, 1$, and let $RR_v \stackrel{\text{def}}{=} E[Y^{a=1} | V = v] / p_v$ be the causal risk ratio in stratum $V = v$. If V is a qualitative effect modifier (Definition 1.2), then V modifies that effect on both the additive and the multiplicative scale (Definition 1.1).*

i Proof

Proof. In each stratum,

$$\begin{aligned} RR_v - 1 &= \frac{E[Y^{a=1} | V = v] - p_v}{p_v} \\ &= \frac{RD_v}{p_v}. \end{aligned} \tag{1}$$

Because $p_v > 0$, $RR_v - 1$ has the same sign as RD_v . Under qualitative modification, RD_1 and RD_0 are nonzero with opposite signs, so $RD_1 \neq RD_0$: additive modification. By Equation 1, $RR_1 - 1$ and $RR_0 - 1$ are also nonzero with opposite signs, so $RR_1 \neq RR_0$: multiplicative modification. $\square$

Without qualitative effect modification, there can be modification on one scale but not the other, as the next example shows.

Example 1.3 (Multiplicative but not additive effect modification). In a second study:

	$\Pr[Y^{a=0} = 1 \mid V = v]$	$\Pr[Y^{a=1} = 1 \mid V = v]$	risk difference	risk ratio
$V = 1$	0.8	0.9	$0.9 - 0.8 = 0.1$	$0.9/0.8 = 1.125$
$V = 0$	0.1	0.2	$0.2 - 0.1 = 0.1$	$0.2/0.1 = 2$

The risk differences are equal (no additive effect modification), but the risk ratios differ (multiplicative effect modification).

Because “effect modification” means nothing without naming the effect measure, some authors say **effect-measure modification** instead.

2 4.2 Stratification to Identify Effect Modification (pp. 49-51)

Definition 2.1 (Stratified Analysis). A **stratified analysis** of the effect of A on Y by a variable V computes that effect separately within each level (**stratum**) $V = v$. For dichotomous V and a dichotomous outcome Y , the stratified causal risk differences are

$$\Pr[Y^{a=1} = 1 \mid V = 1] - \Pr[Y^{a=0} = 1 \mid V = 1] \quad \text{and} \quad \Pr[Y^{a=1} = 1 \mid V = 0] - \Pr[Y^{a=0} = 1 \mid V = 0]. \quad (2)$$

Stratified analysis is the standard tool for detecting effect modification by measured variables.

In real data we see A and Y , not $Y^{a=1}$ and $Y^{a=0}$. How we stratify depends on the design.

2.1 Marginally randomized experiments

If treatment is randomly and unconditionally assigned, exchangeability is expected in **every subset** of the population.

Proposition 2.1 (Stratum-Specific Association Under Marginal Randomization). *Let V be a pre-treatment variable and Y a dichotomous outcome. Suppose that*

- *treatment is assigned independently of the counterfactual outcomes and of V : $(Y^{a=0}, Y^{a=1}, V) \perp\!\!\!\perp A$;*
- *consistency holds: $Y = Y^a$ for individuals with $A = a$;*
- *$\Pr[A = a, V = v] > 0$ for $a = 0, 1$ and every v with $\Pr[V = v] > 0$.*

Then $\Pr[Y = 1 \mid A = a, V = v] = \Pr[Y^a = 1 \mid V = v]$ for each a and v , so the associational risk difference in stratum $V = v$ equals the causal risk difference in that stratum.

i Proof

Proof. The joint independence $(Y^a, V) \perp\!\!\!\perp A$ implies $Y^a \perp\!\!\!\perp A \mid V$. Hence

$$\begin{aligned} \Pr[Y = 1 \mid A = a, V = v] &= \Pr[Y^a = 1 \mid A = a, V = v] && \text{(consistency)} \\ &= \Pr[Y^a = 1 \mid V = v] && (Y^a \perp\!\!\!\perp A \mid V). \end{aligned} \quad (3)$$

Positivity makes every conditional probability on the left well defined. $\square$

So a stratified analysis of association measures identifies effect modification in a marginally randomized experiment.

Stratification yields average effects in subsets of the population, not individual effects (Fine Points 2.1 and 3.2).

2.2 Conditionally randomized experiments: Greeks and Romans

A population of 40: transplant A randomly assigned with probability 0.75 if in severe condition ($L = 1$) and 0.50 otherwise ($L = 0$). Nationality V : 20 Greeks ($V = 1$, data in Table 3.1) and 20 Romans ($V = 0$, Table 4.2).

Table 4.2 (Romans, $V = 0$)

Name	L	A	Y	Name	L	A	Y
Cybele	0	0	0	Latona	1	0	0
Saturn	0	0	1	Mars	1	1	1
Ceres	0	0	0	Minerva	1	1	1
Pluto	0	0	0	Vulcan	1	1	1
Vesta	0	1	0	Venus	1	1	1
Neptune	0	1	0	Seneca	1	1	1
Juno	0	1	1	Proserpina	1	1	1
Jupiter	0	1	1	Mercury	1	1	0
Diana	1	0	0	Juventas	1	1	0
Phoebus	1	0	1	Bacchus	1	1	0

Example 2.1 (The Effect in All 40 Greeks and Romans). Randomization depended only on L , so $Y^a \perp\!\!\!\perp A \mid L$ holds in the combined population, and every stratum of L has treated and untreated individuals. Under consistency, standardization then gives the counterfactual risks. Pooling Greeks and Romans, 16 individuals have $L = 0$ and 24 have $L = 1$, so $\Pr[L = 0] = 0.4$ and $\Pr[L = 1] = 0.6$. The stratum-specific risks are $\Pr[Y = 1 \mid A = 1, L = 0] = 3/8$, $\Pr[Y = 1 \mid A = 1, L = 1] = 12/18 = 2/3$, $\Pr[Y = 1 \mid A = 0, L = 0] = 2/8$, and $\Pr[Y = 1 \mid A = 0, L = 1] = 3/6$. Standardizing by L ,

$$\begin{aligned}
 \Pr[Y^{a=1} = 1] &= (3/8)(0.4) + (2/3)(0.6) \\
 &= 0.15 + 0.40 \\
 &= 0.55, \\
 \Pr[Y^{a=0} = 1] &= (2/8)(0.4) + (3/6)(0.6) \\
 &= 0.10 + 0.30 \\
 &= 0.40.
 \end{aligned} \tag{4}$$

The causal risk difference in the population of 40 is 0.15 and the causal risk ratio is $0.55/0.40 = 1.375$.

The same calculation applies to observational data if we are willing to assume $Y^a \perp\!\!\!\perp A \mid L$ (Chapter 3). The book states these two risks without the intermediate steps.

To compute $\Pr[Y^a = 1 \mid V = v]$ in each stratum, stratify first and adjust second.

Algorithm 2.1 (Stratify by the Modifier, Then Adjust for the Confounders). *Let V be a pre-treatment effect modifier of interest and L a set of measured pre-treatment covariates. Assume consistency, conditional exchangeability $Y^a \perp\!\!\!\perp A \mid L, V$, and positivity $\Pr[A = a \mid L = l, V = v] > 0$ whenever $\Pr[L = l, V = v] > 0$.*

1. **Stratify** by V .
2. Within each stratum $V = v$, **standardize** by L :

$$\Pr[Y^a = 1 \mid V = v] = \sum_l \Pr[Y = 1 \mid A = a, L = l, V = v] \Pr[L = l \mid V = v], \tag{5}$$

or, equivalently, IP weight with weights that depend on L (and V).

3. Compare the resulting stratum-specific causal risk differences or risk ratios across levels of V .

Step 2 can be skipped when $Y^a \perp\!\!\!\perp A \mid V$ already holds and $\Pr[A = a \mid V = v] > 0$, that is, when V itself suffices for conditional exchangeability (Section 4.4).

In the Greeks and Romans example, randomization depended only on L , so A is independent of $(Y^{a=0}, Y^{a=1}, V)$ given L , and in particular $Y^a \perp\!\!\!\perp A \mid L, V$.

Example 2.2 (Effect modification by nationality). Apply Algorithm 2.1 with $V = \text{nationality}$.

Greeks ($V = 1$): as computed in Chapter 2, causal risk difference 0, risk ratio 1.

Romans ($V = 0$): from Table 4.2, 8 Romans have $L = 0$ and 12 have $L = 1$, so $\Pr[L = 0 \mid V = 0] = 0.4$ and $\Pr[L = 1 \mid V = 0] = 0.6$, and

$$\begin{aligned}\Pr[Y^{a=1} = 1 \mid V = 0] &= \Pr[Y = 1 \mid A = 1, L = 0, V = 0](0.4) + \Pr[Y = 1 \mid A = 1, L = 1, V = 0](0.6) \\ &= (2/4)(0.4) + (6/9)(0.6) \\ &= 0.2 + 0.4 \\ &= 0.6,\end{aligned}$$

$$\begin{aligned}\Pr[Y^{a=0} = 1 \mid V = 0] &= \Pr[Y = 1 \mid A = 0, L = 0, V = 0](0.4) + \Pr[Y = 1 \mid A = 0, L = 1, V = 0](0.6) \\ &= (1/4)(0.4) + (1/3)(0.6) \\ &= 0.1 + 0.2 \\ &= 0.3.\end{aligned}$$

Causal risk difference $0.6 - 0.3 = 0.3$, risk ratio $0.6/0.3 = 2$.

The effects differ between strata, so nationality modifies the effect on both the additive and multiplicative scales. The modification is **not qualitative**: the effect is harmful or null in both strata.

2.3 Surrogate and causal effect modifiers

Nationality may only be a **marker** for the factor truly responsible.

Definition 2.2 (Modifiers That May Be Only Markers). Suppose there is effect modification by V (Definition 1.1). V is a **surrogate effect modifier** when the effect varies across levels of V only because V is associated with another variable W whose levels drive that variation, so that an intervention that changed W while leaving V alone could remove the modification by V . Such a variable W is a **causal effect modifier**. This is a working definition for these notes; Section 6.6 makes both kinds precise with causal diagrams.

Example 2.3 (Nationality and Quality of Care). Suppose Greek hospitals provide better heart surgery than Roman ones. Then the modification by nationality in Example 2.2 could be due to quality of care: an intervention that improved surgery in Rome could remove the modification by passport-defined nationality. Nationality would be a surrogate effect modifier, and quality of care a causal effect modifier.

⚠ Modification Does Not Make the Modifier a Cause

“Effect modification by V ” does not imply that V plays a causal role; some authors prefer the more neutral “effect heterogeneity across strata of V ”. Chapter 5’s **interaction** does attribute a causal role to the variables involved.

i Fine Point 4.1: Effect in the Treated

The individuals who actually received treatment form one subset of special interest. Definition 2.3 gives the average causal effect in the treated and in the untreated, and Remark 2.1 explains when the effect in the treated differs from the population effect (Hernán and Robins 2020, 50).

Definition 2.3 (Average Causal Effect in the Treated and in the Untreated). Let A and Y be dichotomous, with $\Pr[A = 1] > 0$. Among the treated, the **causal risk difference in the treated** equals $\Pr[Y^{a=1} = 1 \mid A = 1] - \Pr[Y^{a=0} = 1 \mid A = 1]$, and the **causal risk ratio in the treated** is $\Pr[Y^{a=1} = 1 \mid A = 1] / \Pr[Y^{a=0} = 1 \mid A = 1]$. The latter, defined when $\Pr[Y^{a=0} = 1 \mid A = 1] > 0$, is also called the **standardized morbidity ratio (SMR)**. Replacing $A = 1$ by $A = 0$ in the conditioning event (with $\Pr[A = 0] > 0$, and a positive denominator for the ratio) gives the causal risk difference and ratio **in the untreated** (book Figure 4.1).

Under consistency, $\Pr[Y^{a=1} = 1 \mid A = 1] = \Pr[Y = 1 \mid A = 1]$, so the treated have a non-null average causal effect exactly when

$$\Pr[Y = 1 \mid A = 1] \neq \Pr[Y^{a=0} = 1 \mid A = 1]. \quad (6)$$

Remark 2.1 (When the Effect in the Treated Differs from the Population Effect). If individual causal effects are distributed differently among the treated than among the untreated, the effect in the treated generally differs from the population effect. Treatment group then acts as a marker for the true modifiers, but we would not say the effect of A is “modified by A ”; the modification is by unidentified variables distributed differently between treatment groups. The book focuses on population effects because effects in the treated or untreated do not generalize directly to time-varying treatments (Part III).

3 4.3 Why Care About Effect Modification (pp. 51-53)

Reasons to identify effect modification, and to collect pre-treatment descriptors V even in randomized experiments:

1. the average effect **differs between populations** with different prevalence of V (transportability, defined below);
2. it identifies **who would benefit most** from an intervention;
3. it may help understand **mechanisms**.

3.1 1. Populations with different mixes of modifiers

In Table 4.1 the effect is harmful in women and beneficial in men, and the population effect is null only because each sex is 50%. In a population with more women (e.g., graduating college students), the average effect would be harmful.

- With non-qualitative effect modification, the **magnitude** (but not the direction) of the average effect varies across populations, e.g., the effects of asbestos exposure (different in smokers and nonsmokers) or of universal health care (different in low- and high-income families).

! Name the Population Along with the Effect

There is no such thing as “the average causal effect of A on Y (period)”, only “the average causal effect of A on Y in a population with a particular mix of causal effect modifiers.”

3.2 Transportability

Definition 3.1 (Transporting an Effect Between Populations). **Transportability** is the extrapolation of an average causal effect from the population in which it was computed (the **study population**) to another population (the **target population**). Lack of transportability is sometimes called lack of external validity.

The effect in the population of Greeks and Romans may not transport to a population with a different mix of sex and nationality.

- Conditional effects within strata of modifiers may transport better, but there is no guarantee: other unmeasured modifiers may be distributed differently.
- Those unmeasured modifiers are not variables needed for exchangeability; they are just risk factors for the outcome, about which we have much less information.

! Transportability Cannot Be Checked in the Data

Transportability is harder to justify than identification within one population. It is an **unverifiable assumption** that rests on subject-matter knowledge.

The book’s examples: most experts would agree that the health effects of raising a household’s annual income by \$100 in Niger cannot be transported to the Netherlands, but that the effects of cholesterol-lowering drugs in Europeans can be transported to Canadians. Smith and Pell (2003) found no major modifiers of the effect of parachute use on death after “gravitational challenge”, so trials restricted to one group would not compromise transportability. The book cites Hernán and VanderWeele (2011), Pearl and Bareinboim (2014), and Dahabreh and Hernán (2019) on effect modification and transportability (Hernán and Robins 2020, 52).

i Fine Point 4.2: Transportability

Whether an effect carries over from a study population to a target population depends on features of both. Remark 3.1 lists the features whose similarity can justify transporting an effect, and Remark 3.2 shows how stratum-specific effects can be reweighted to a target population, and why that may still fall short (Hernán and Robins 2020, 54). Methods for transporting effects are discussed by Westreich et al. (2017), Rudolph and van der Laan (2017), and Dahabreh et al. (2020b).

Remark 3.1 (Similarities That Support Transporting an Effect). An average causal effect (Definition 3.1) may be transported from a study population to a target population if the two populations are similar with respect to:

- **effect modification**: the distribution of effect modifiers;
- **the definition of treatment**: different versions of treatment can have different effects;
- **interference** (Fine Point 1.1): e.g., a socially active person may bring friends along to exercise, so contact patterns affect the magnitude of an exercise intervention’s effect.

Remark 3.2 (Reweighting Strata to a Target Population). Let S index strata defined by measured effect modifiers (e.g., Roman women, Greek women, Roman men, Greek men), let $\Pr$

and $\Pr^*$ denote probabilities in the study and the target population, and let $q_s \stackrel{\text{def}}{=} \Pr^*[S = s]$. Suppose every stratum with $q_s > 0$ is also present in the study population, and, for each a and s , the stratum-specific counterfactual risk in the target population equals the one computed in the study population: $\Pr^*[Y^a = 1 \mid S = s] = \Pr[Y^a = 1 \mid S = s]$. Then the target population's counterfactual risk under treatment level a is

$$\Pr^*[Y^a = 1] = \sum_s \Pr[Y^a = 1 \mid S = s] q_s, \quad (7)$$

and the target causal risk difference and risk ratio (when $\Pr^*[Y^{a=0} = 1] > 0$) follow by contrasting these risks for $a = 1$ and $a = 0$. The assumption that the stratum-specific risks carry over can fail because of unmeasured effect modifiers, different interference patterns, or different versions of treatment, so the reweighted value need not equal the true effect in the target population.

i Technical Point 4.1: Computing the Effect in the Treated

Effects among the treated, or among the untreated, need a weaker exchangeability condition than the population effect. Definition 3.2 names that condition, and Proposition 3.1 shows how standardization and IP weighting compute the needed counterfactual means; Sato and Matsuyama (2003) derived the IP weighting equality for dichotomous A . Miettinen (1972) and Greenland and Rothman (2008) discuss standardized risk ratios, and Hernán and Robins (2006a) give further details (Hernán and Robins 2020, 52).

Definition 3.2 (Partial Exchangeability). Let L be a set of measured pre-treatment covariates. For a treatment level a , **partial exchangeability** is $Y^a \perp\!\!\!\perp A \mid L$ for that one level, whereas the population effect uses conditional exchangeability for both $a = 0$ and $a = 1$. The effect in the treated (Definition 2.3) uses partial exchangeability for $a = 0$, and the effect in the untreated uses it for $a = 1$. For the effect in the treated, for example, it does not matter whether the untreated would have had the same risk as the treated had they been treated.

Proposition 3.1 (Identifying Counterfactual Means in a Treatment Group). *Let L be a set of measured pre-treatment covariates, and let $a, a' \in \{0, 1\}$ with $\Pr[A = a'] > 0$. Assume consistency, partial exchangeability $Y^a \perp\!\!\!\perp A \mid L$ for this a (Definition 3.2), and positivity $f(a \mid l) > 0$ for every l with $\Pr[L = l] > 0$, where $f(a' \mid l) \stackrel{\text{def}}{=} \Pr[A = a' \mid L = l]$ is the conditional probability mass function of A given L . Then $E[Y^a \mid A = a']$ equals both the standardized mean*

$$E[Y^a \mid A = a'] = \sum_l E[Y \mid A = a, L = l] \Pr[L = l \mid A = a'] \quad (8)$$

and the IP weighted mean with weights $\Pr[A = a' \mid L]/f(A \mid L)$:

$$E[Y^a \mid A = a'] = \frac{E\left[\frac{\mathbb{1}(A = a) Y}{f(A \mid L)} \Pr[A = a' \mid L]\right]}{E\left[\frac{\mathbb{1}(A = a)}{f(A \mid L)} \Pr[A = a' \mid L]\right]}. \quad (9)$$

i Proof

Proof. Standardization. For each l with $\Pr[L = l \mid A = a'] > 0$,

$$\begin{aligned}
E[Y^a \mid A = a', L = l] &= E[Y^a \mid L = l] && \text{(partial exchangeability)} \\
&= E[Y^a \mid A = a, L = l] && \text{(partial exchangeability; positivity)} \\
&= E[Y \mid A = a, L = l] && \text{(consistency)}.
\end{aligned} \tag{10}$$

Averaging over $\Pr[L = l \mid A = a']$ (law of total expectation) gives Equation 8.

IP weighting. Because $\mathbb{1}(A = a)$ is zero unless $A = a$, the numerator of Equation 9 is

$$\begin{aligned}
E\left[\frac{\mathbb{1}(A = a)Y}{f(A \mid L)} \Pr[A = a' \mid L]\right] &= E\left[\frac{\Pr[A = a \mid L] E[Y \mid A = a, L]}{f(a \mid L)} \Pr[A = a' \mid L]\right] && \text{(iterated expectations given } L) \\
&= E[E[Y \mid A = a, L] \Pr[A = a' \mid L]] && \text{(definition of } f; \text{ positivity)} \\
&= \sum_l E[Y \mid A = a, L = l] \Pr[L = l \mid A = a'] \Pr[A = a'] && \text{(Bayes' rule)}.
\end{aligned} \tag{11}$$

The same steps with Y replaced by 1 show that the denominator equals $\Pr[A = a']$. Their ratio is the standardized mean, which equals $E[Y^a \mid A = a']$ by Equation 8. $\square$

3.3 2. Who benefits most

If physicians knew of the qualitative modification by sex in Table 4.1, then, lacking other information, they would treat the next patient only if he is a man.

In the second study (Example 1.3), treatment changes the risk by the same 0.1 in both strata, so treating everyone would change risk equally in both strata, despite the multiplicative effect modification.

Remark 3.3 (Use the Additive Scale to Target Treatment). To identify the groups that would benefit most from an intervention, look for **additive**, not multiplicative, effect modification: the causal risk difference in a stratum is the average change in risk that treating everyone in that stratum would produce, whatever the risk ratio in that stratum is.

Proposition 3.2 (Modification on at Least One Scale). *Let V be binary with $\Pr[V = v] > 0$ for $v = 0, 1$, and let $p_v \stackrel{\text{def}}{=} \Pr[Y^{a=0} = 1 \mid V = v] > 0$, $\text{RD}_v \stackrel{\text{def}}{=} \Pr[Y^{a=1} = 1 \mid V = v] - p_v$, and $\text{RR}_v \stackrel{\text{def}}{=} \Pr[Y^{a=1} = 1 \mid V = v]/p_v$ for $v = 0, 1$. If $\text{RD}_v \neq 0$ for at least one v and $p_1 \neq p_0$, then there is effect modification by V on the additive scale, the multiplicative scale, or both.*

Proof

Proof. Suppose there is no additive modification, so $\text{RD}_1 = \text{RD}_0$; write d for this common value. Since at least one RD_v is nonzero, $d \neq 0$. By the definitions, $\Pr[Y^{a=1} = 1 \mid V = v] = p_v + d$, so $\text{RR}_v = 1 + d/p_v$. Because $d \neq 0$ and $p_1 \neq p_0$, $d/p_1 \neq d/p_0$, so $\text{RR}_1 \neq \text{RR}_0$: there is multiplicative modification. $\square$

In Example 1.3, both strata have causal risk difference 0.1, while $\Pr[Y^{a=0} = 1 \mid V = 1] = 0.8$ and $\Pr[Y^{a=0} = 1 \mid V = 0] = 0.1$, so the risk ratios are $1 + 0.1/0.8 = 1.125$ and $1 + 0.1/0.1 = 2$, as Proposition 3.2 predicts.

Report the Counterfactual Risks

Reporting both counterfactual risks, $\Pr[Y^{a=1} = 1 \mid V = v]$ and $\Pr[Y^{a=0} = 1 \mid V = v]$, for every stratum tells the reader more than reporting only their difference or ratio: the risks

determine both scales, and neither scale alone determines the risks.

Several authors (e.g., Blot and Day 1979; Rothman et al. 1980; Saracci 1980) have called additive effect modification the scale of interest for public health.

In the second study, multiplicative modification is expected: with $\Pr[Y^{a=0} = 1 \mid V = 1] = 0.8$, the causal risk ratio in that stratum can be at most $1/0.8 = 1.25$, so it must differ from the risk ratio of 2 in the $V = 0$ stratum. Multiplicative modification here arises from differences in the untreated risk across levels of V .

(The book’s text says that treatment “reduces” the risk by 10% in this example; in its numbers, 0.8 to 0.9 and 0.1 to 0.2, treatment increases the risk by 0.1 in both strata. The argument is unaffected.)

3.4 3. Mechanisms

Effect modification may give clues about biological, social, or other mechanisms: for example, a greater risk of HIV infection in uncircumcised compared with circumcised men. It can also be a first step towards characterizing interactions between two treatments, a related but distinct causal concept (Chapter 5).

4 4.4 Stratification as a Form of Adjustment (pp. 53-55)

Two different goals, two different tools:

- **standardization** (or IP weighting) adjusts for the variables L needed for conditional exchangeability;
- **stratification** by V identifies effect modification.

In practice, stratification is often used *instead of* standardization to adjust for L , so much so that many investigators treat “stratification” and “adjustment” as synonyms.

Example 4.1 (Stratification as adjustment in Table 2.2). Asked to “adjust for L ”, an epidemiologist splits Table 2.2 (the same data as Table 3.1) into $L = 0$ and $L = 1$ and reports

$$\Pr[Y = 1 \mid A = 1, L = l] / \Pr[Y = 1 \mid A = 0, L = l] = 1 \quad \text{for } l = 0, 1$$

($1/4 \div 1/4$ for $l = 0$ and $2/3 \div 2/3$ for $l = 1$). Under $Y^a \perp\!\!\!\perp A \mid L$, each is the causal risk ratio in its stratum, because then $\Pr[Y = 1 \mid A = a, L = l] = \Pr[Y^a = 1 \mid L = l]$.

4.1 Conditional versus marginal effect measures

Definition 4.1 (Effect Measures Within Strata and in the Whole Population). Let L be a pre-treatment variable. A **conditional** effect measure contrasts counterfactual outcomes within the subset of the population with $L = l$, e.g., the conditional causal risk ratio $\Pr[Y^{a=1} = 1 \mid L = l] / \Pr[Y^{a=0} = 1 \mid L = l]$. A **marginal** (unconditional) effect measure contrasts them in the whole population, e.g., the marginal causal risk ratio $\Pr[Y^{a=1} = 1] / \Pr[Y^{a=0} = 1]$.

- The two stratum-specific risk ratios in Example 4.1 are **conditional** effect measures; the risk ratio of 1 from Chapter 2 is a **marginal** one. All three equal 1 here only because L does not modify the effect.
- Each stratum-specific measure describes a nonoverlapping subset; in general none of them is the population effect. That is why Chapter 2 used standardization and IP weighting rather than stratification.

- Stratification as adjustment must be done within **every combination** of the variables in L : Romans with $L = 1$, Greeks with $L = 1$, Romans with $L = 0$, Greeks with $L = 0$. The association in the stratum $V = 0$ alone does not give the effect in Romans, because V by itself does not guarantee conditional exchangeability.

So stratification forces one to evaluate effect modification by all variables in L , whether or not that is of interest. Stratifying by V and then standardizing or IP weighting for L handles exchangeability and effect modification separately.

⚠ More Pitfalls of Stratification as Adjustment

- Some effect measures, such as the odds ratio, are **noncollapsible**: their population value need not be a weighted average of their stratum-specific values (Fine Point 4.3).
- With time-varying treatments, when exchangeability requires adjusting for time-varying L affected by prior treatment, stratifying on L is inappropriate adjustment (Part III): stratum-specific measures may then lack a causal interpretation even under exchangeability, positivity, and well-defined interventions (Robins 1986, 1987; Chapter 20).
- Stratification requires **positivity** as well as exchangeability: no effect can be computed in a stratum $L = l$ containing only treated or only untreated individuals.

4.2 Restriction

Definition 4.2 (Restricting the Analysis to Some Strata). Let L be a set of measured pre-treatment covariates. **Restriction** computes the effect of A on Y only in some strata of L , for example only in $L = 1$. Stratification is restriction applied to each of several mutually exclusive subsets that together cover the population, with conditional exchangeability required in each.

When positivity fails in some strata, restriction limits the inference to the strata where positivity holds (Chapter 3).

5 4.5 Matching as Another Form of Adjustment (pp. 55-56)

Definition 5.1 (Matching). **Matching** selects individuals so that, among those selected, a set of measured pre-treatment variables L (the **matching factors**) is distributed identically in the treated and the untreated.

- In one-to-one matching, each individual in a reference group (e.g., the untreated) is paired with an individual of the other treatment group who has the same value of L , forming **matched pairs**; one-to-many matching forms **matched sets**.
- **Frequency matching** instead selects individuals in each treatment group so that L has a chosen distribution in both groups, e.g., selecting treated individuals so that 70% have $L = 1$, then doing the same for the untreated.

Example 5.1 (Matching in Table 2.2). For each untreated individual, randomly select a treated individual with the same L (the matching factor). One possible set of 7 matched pairs (untreated first):

- $L = 0$: Rheia-Hestia, Kronos-Poseidon, Demeter-Hera, Hades-Zeus
- $L = 1$: Artemis-Ares, Apollo-Aphrodite, Leto-Hermes

All untreated but only some treated are selected. In the matched population, the proportion with $L = 1$ is 3/7 in both groups by design. The risk is 3/7 in the treated (Zeus, Ares, Aphrodite died) and 3/7 in the untreated (Kronos, Artemis, Apollo died), so the causal risk

ratio is 1.

Under conditional exchangeability given L , replacing the treated by a subset with the same distribution of L as the untreated yields (unconditional) exchangeability in the matched population, so average outcomes can be compared directly. Matching also ensures positivity in the matched population, because strata containing only treated or only untreated individuals are excluded.

This discussion applies to cohort studies. In case-control designs (Chapter 8), cases and non-cases are matched; even if the matching factors suffice for conditional exchangeability, this does not produce unconditional exchangeability of treated and untreated, and adjusting for the matching factors via stratification is needed to estimate conditional effect measures.

5.1 Which population does matching describe?

Remark 5.1 (The Reference Group Sets the Target Population). In matching (Definition 5.1), the group used as the reference for finding matches defines the population in which the effect is computed. Matching each untreated individual to a treated one, as in Example 5.1, estimates the effect in the untreated (Definition 2.3); with fewer treated than untreated in every stratum of L , one usually matches each treated individual and estimates the effect in the treated.

- With several variables in L , matches come from the same stratum of the combination of all of them.

Because the matched population is a subset of the original one, its distribution of effect modifiers generally differs, which is the topic of Section 4.6.

As the number of matching factors grows, so does the chance that no exact match exists. The literature on approximate matching is beyond the book's scope; Stuart (2010) is an introduction.

6 4.6 Effect Modification and Adjustment Methods (pp. 56-60)

Remark 6.1 (What Each Adjustment Method Computes). Standardization, IP weighting, stratification/restriction, and matching compute **different types** of causal effects (Definition 4.1):

- standardization and IP weighting can compute **marginal or conditional** effects;
- stratification/restriction and matching compute only **conditional** effects in certain subsets.

All four require exchangeability and positivity, but only in the subset to which the effect refers: for the effect among $L = l$, only in that subset; for the marginal effect, in all levels of L .

Without effect modification all four give the same answer: in Table 2.2 the causal risk ratio was 1 in the population, in each stratum of L , and in the untreated.

6.1 Table 4.3: four correct answers

Outcome Z = high blood pressure. Assume $Z^a \perp\!\!\!\perp A \mid L$ and positivity.

Name	L	A	Z	Name	L	A	Z
Rheia	0	0	0	Leto	1	0	0
Kronos	0	0	1	Ares	1	1	1

Name	L	A	Z	Name	L	A	Z
Demeter	0	0	0	Athena	1	1	1
Hades	0	0	0	Hephaestus	1	1	1
Hestia	0	1	0	Aphrodite	1	1	0
Poseidon	0	1	0	Polyphemus	1	1	0
Hera	0	1	1	Persephone	1	1	0
Zeus	0	1	1	Hermes	1	1	0
Artemis	1	0	1	Hebe	1	1	0
Apollo	1	0	1	Dionysus	1	1	0

Example 6.1 (Four causal risk ratios from Table 4.3). Stratum-specific risks: $\Pr[Z = 1 \mid A = 1, L = 0] = 2/4$, $\Pr[Z = 1 \mid A = 0, L = 0] = 1/4$, $\Pr[Z = 1 \mid A = 1, L = 1] = 3/9 = 1/3$, $\Pr[Z = 1 \mid A = 0, L = 1] = 2/3$; $\Pr[L = 0] = 0.4$.

- **Stratification:** risk ratio $\frac{2/4}{1/4} = 2.0$ in $L = 0$ and $\frac{1/3}{2/3} = 0.5$ in $L = 1$.
- **Standardization / IP weighting** (population):

$$\begin{aligned}
\Pr[Z^{a=1} = 1] &= (2/4)(0.4) + (1/3)(0.6) \\
&= 0.2 + 0.2 \\
&= 0.4, \\
\Pr[Z^{a=0} = 1] &= (1/4)(0.4) + (2/3)(0.6) \\
&= 0.1 + 0.4 \\
&= 0.5,
\end{aligned}$$

so the risk ratio is $0.4/0.5 = 0.8$.

- **Matching** (the matched pairs of Section 4.5; effect in the untreated): $\Pr[Z^{a=1} = 1 \mid A = 0] / \Pr[Z = 1 \mid A = 0] = (3/7)/(3/7) = 1.0$.

Four different numbers, 0.8, 2.0, 0.5, and 1.0, and all of them are correct.

The explanation (leaving aside random variability) is **qualitative effect modification** by L : treatment doubles the risk when $L = 0$ and halves it when $L = 1$.

- The population effect (0.8) is beneficial because the ratio $\Pr[Z^{a=0} = 1 \mid L = 1] / \Pr[Z^{a=0} = 1 \mid L = 0] = (2/3)/(1/4) = 8/3$ exceeds 2 times the odds of being noncritical, $2 \Pr[L = 0] / \Pr[L = 1] = 2(0.4/0.6) = 4/3$ (Technical Point 4.3).
- The effect in the untreated is null (1.0) because the untreated contain a larger proportion of noncritical individuals ($4/7$) than the population (0.4).

The lesson: always specify the population, or subset, to which an effect measure corresponds.

Proposition 6.1 (When is the marginal risk ratio below 1? (Technical Point 4.3)). *Let $\text{RR}_l \stackrel{\text{def}}{=} \Pr[Z^{a=1} = 1 \mid L = l] / \Pr[Z^{a=0} = 1 \mid L = l]$, $p_l \stackrel{\text{def}}{=} \Pr[Z^{a=0} = 1 \mid L = l]$, and $\pi_l \stackrel{\text{def}}{=} \Pr[L = l]$. With $\text{RR}_0 = 2$ and $\text{RR}_1 = 0.5$, the marginal risk ratio is below 1 if and only if*

$$\frac{p_1}{p_0} > \frac{2\pi_0}{\pi_1}.$$

i Proof

Proof. The marginal risk ratio is a weighted average of the conditional ones:

$$\begin{aligned}
\frac{\Pr[Z^{a=1} = 1]}{\Pr[Z^{a=0} = 1]} &= \frac{\sum_l \Pr[Z^{a=1} = 1 \mid L = l] \pi_l}{\Pr[Z^{a=0} = 1]} && \text{(law of total probability)} \\
&= \sum_l \text{RR}_l \frac{p_l \pi_l}{\Pr[Z^{a=0} = 1]} && \text{(multiply and divide by } p_l) \\
&= \sum_l \text{RR}_l w(l), && w(l) \stackrel{\text{def}}{=} \frac{p_l \pi_l}{\Pr[Z^{a=0} = 1]}, \sum_l w(l) = 1.
\end{aligned}$$

Substituting $\text{RR}_0 = 2$, $\text{RR}_1 = 0.5$, and $w(1) = 1 - w(0)$:

$$\begin{aligned}
2w(0) + 0.5(1 - w(0)) < 1 &\iff 1.5w(0) < 0.5 \\
&\iff w(0) < 1/3 \\
&\iff 3p_0\pi_0 < p_0\pi_0 + p_1\pi_1 && \text{(definition of } w(0); \Pr[Z^{a=0} = 1] = p_0\pi_0 + p_1\pi_1) \\
&\iff 2p_0\pi_0 < p_1\pi_1 \\
&\iff \frac{p_1}{p_0} > \frac{2\pi_0}{\pi_1}.
\end{aligned}$$

In Table 4.3, $w(0) = (1/4)(0.4)/0.5 = 0.2 < 1/3$, and indeed $2(0.2) + 0.5(0.8) = 0.8$. $\square$

6.2 A well-characterized target population

Remark 6.2 (A Second Prerequisite for Causal Inference). Chapter 3 required the causal effect of interest to be well defined before it can be estimated. A **well-characterized target population** is a second requirement of that kind: the population, or subset, to which the effect refers must be stated. Experiments that compare interventions in a population meeting a priori eligibility criteria have both automatically; observational studies must define them explicitly.

Example 6.2 (The Effect in the Untreated in Table 4.3 by Standardization). Under $Z^{a=1} \perp\!\!\!\perp A \mid L$, Proposition 3.1 with $a = 1$ and $a' = 0$ computes the effect in the untreated without matching. Of the 7 untreated individuals in Table 4.3, 4 have $L = 0$ and 3 have $L = 1$, so

$$\begin{aligned}
\Pr[Z^{a=1} = 1 \mid A = 0] &= \Pr[Z = 1 \mid A = 1, L = 0] \Pr[L = 0 \mid A = 0] + \Pr[Z = 1 \mid A = 1, L = 1] \Pr[L = 1 \mid A = 0] \\
&= (2/4)(4/7) + (1/3)(3/7) \\
&= 3/7.
\end{aligned} \tag{12}$$

By consistency, $\Pr[Z^{a=0} = 1 \mid A = 0] = \Pr[Z = 1 \mid A = 0] = 3/7$, so the causal risk ratio in the untreated is 1.0, the same answer that matching gave in Example 6.1.

An investigator who used IP weighting (population effect) and one who used matching (effect in the untreated) need not debate which method is better: their answers differ because they asked different causal questions. The second could also have obtained the effect in the untreated by standardization or IP weighting, as in Example 6.2.

Part II shows how standardization, IP weighting, and stratification combine with parametric or semiparametric models. Standard regression models are a form of stratification: the treatment-outcome association is estimated within levels of all the other covariates in the model.

Finally, stratification yields average effects in subsets, not individual effects, which are identified only under extreme assumptions (Fine Points 2.1 and 3.2).

i Technical Point 4.2: Pooling of Stratum-Specific Effect Measures

Until Chapter 10, the book works with whole populations and so talks of computing, not estimating, effects. With samples, stratum-specific estimates are variable, and pooling them is a common response. Definition 6.1 describes pooling, Example 6.3 pools the Table 4.3 risk ratios, and the warning after that example explains when the pooled value is hard to interpret. Greenland and Rothman (2008) review common methods for stratified analysis (Hernán and Robins 2020, 57).

Definition 6.1 (Pooled Stratum-Specific Effect Measure). A **pooled** effect measure combines the stratum-specific estimates of an effect measure across strata of L into a single estimate (e.g., by the Woolf, Mantel-Haenszel, or maximum likelihood method), or comes from a regression model saturated with respect to L with no product terms between A and L . Pooling aims at a narrower confidence interval around a *common* stratum-specific effect; the pooled measure is still a conditional effect measure (Definition 4.1).

Example 6.3 (Mantel-Haenszel Risk Ratio in Table 4.3). In stratum l , let a_l and b_l be the numbers of cases among the treated and the untreated, n_{1l} and n_{0l} the numbers treated and untreated, and N_l the stratum size. In Table 4.3, the stratum-specific risk ratios for Z are 2.0 ($L = 0$) and 0.5 ($L = 1$), and the Mantel-Haenszel risk ratio is

$$\begin{aligned} \text{RR}_{\text{MH}} &= \frac{\sum_l a_l n_{0l} / N_l}{\sum_l b_l n_{1l} / N_l} \\ &= \frac{(2 \cdot 4)/8 + (3 \cdot 3)/12}{(1 \cdot 4)/8 + (2 \cdot 9)/12} \\ &= \frac{1.75}{2} \\ &\approx 0.88. \end{aligned} \tag{13}$$

A Pooling Assumes a Common Effect

The pooled value in Example 6.3 is meaningful only if the stratum-specific estimates 2.0 and 0.5 estimate the same effect (e.g., a true risk ratio of 0.9 in both strata, with small-sample noise); if the true stratum effects really are 0.5 and 2.0, the pooled value is hard to interpret. The same issue arises in meta-analyses of heterogeneous studies (Dahabreh et al. 2020a). The finer the stratification, the harder it is to tell sampling variability from effect-measure modification.

i Fine Point 4.3: Collapsibility and the Odds Ratio

A desirable property of an effect measure is that its population value is a weighted average of its stratum-specific values, and so lies within their range. Definition 6.2 names that property, and Example 6.4 uses the data in Table 4.4 to show that the odds ratio can lack it (Hernán and Robins 2020, 60).

Definition 6.2 (Collapsible Effect Measure). In a given population, an effect measure is **collapsible** over a variable V when its value in the whole population is a weighted average of its values in the strata of V , with nonnegative weights that sum to 1, so that the population value lies within the range of the stratum-specific values. It is **noncollapsible** in that population otherwise.

In follow-up studies with V not affected by treatment, the risk difference and the risk ratio are collapsible. The causal risk difference is the average of the stratum-specific ones with weights

$\Pr[V = v]$. When $\Pr[Y^{a=0} = 1 \mid V = v] > 0$ in every stratum, the causal risk ratio is the average of the stratum-specific ones with weights $\Pr[Y^{a=0} = 1 \mid V = v] \Pr[V = v] / \Pr[Y^{a=0} = 1]$. The odds ratio (and the rarely used odds difference) is not collapsible in general (Greenland 1987).

Table 4.4: effect of moving to high altitude A (top of Mount Olympus) on depression Y ; $V = 1$ for women. The decision to move was random, so $Y^a \perp\!\!\!\perp A$.

Name	V	A	Y	Name	V	A	Y
Rheia	1	0	0	Kronos	0	0	0
Demeter	1	0	0	Hades	0	0	0
Hestia	1	0	0	Poseidon	0	0	1
Hera	1	0	0	Zeus	0	0	1
Artemis	1	0	1	Apollo	0	0	0
Leto	1	1	0	Ares	0	1	1
Athena	1	1	1	Hephaestus	0	1	1
Aphrodite	1	1	1	Polyphemus	0	1	1
Persephone	1	1	0	Hermes	0	1	0
Hebe	1	1	1	Dionysus	0	1	1

Example 6.4 (The odds ratio is not collapsible).

	risk if $A = 1$	risk if $A = 0$	risk ratio	odds ratio
men ($V = 0$)	4/5	2/5	2	$\frac{4/1}{2/3} = 6$
women ($V = 1$)	3/5	1/5	3	$\frac{3/2}{1/4} = 6$
population	7/10	3/10	$7/3 \approx 2.3$	$\frac{7/3}{3/7} = 49/9 \approx 5.4$

The population risk ratio (2.3) lies between the stratum-specific ones (2 and 3). The population odds ratio (5.4) is closer to the null than **both** stratum-specific odds ratios (6), even though there is no confounding.

Remark 6.3 (Where Noncollapsibility Comes From, and When It Vanishes). Noncollapsibility of the odds ratio is one instance of Jensen’s inequality (Samuels 1981). The population odds ratio may lie nearer the null than every non-null stratum-specific odds ratio when V predicts Y on its own and is independent of A , as in Table 4.4 (Miettinen and Cook 1981). Under the sharp null the odds ratio is collapsible. If, in a follow-up study, fewer than about 10% of individuals in each stratum develop the outcome, the odds ratio is close to collapsible and close to the risk ratio.

A Changed Odds Ratio Does Not Prove Lack of Exchangeability

A difference between the conditional and the unconditional odds ratio does not by itself indicate lack of exchangeability. In Example 6.4 the odds ratio changes by about 10% ($6/5.4 \approx 1.1$) although the treated and untreated are exchangeable. Greenland, Robins, and Pearl (1999) review the relation between noncollapsibility and lack of exchangeability.

7 Summary

- V **modifies** the effect of A on Y when the average causal effect varies across levels of V ; whether it does depends on the effect measure (additive vs. multiplicative). Qualitative modification (opposite directions) appears on both scales.
- In a marginally randomized experiment, stratum-specific associations identify stratum-specific effects; otherwise, **stratify by V , then standardize or IP weight for L** .
- Effect modifiers may be **surrogate** rather than causal.
- Effect modification matters for **transportability**, for deciding **who benefits** (on the additive scale), and for **mechanisms**. Report absolute counterfactual risks in each stratum.
- **Stratification/restriction** and **matching** adjust for L but estimate conditional effects (in strata, or in the treated/untreated); **standardization** and **IP weighting** can estimate marginal effects. Under effect modification these differ, and all can be correct: specify the target population.
- The **odds ratio is noncollapsible**.

Looking ahead:

- Chapter 5 introduces **interaction** between two treatments, which, unlike effect modification, attributes a causal role to both variables
- Chapter 6 introduces causal diagrams; Section 6.6 represents surrogate and causal effect modifiers graphically
- Part II combines standardization, IP weighting, and stratification with models

8 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC.
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